



PYQ 2010–2025

MBBS BIOCHEMISTRY · MODULE 02

Carbohydrate Metabolism

Glycolysis · TCA Cycle · HMP Shunt · Glycogen · Gluconeogenesis

SUBJECT**Biochemistry — Paper 1****EXAM RELEVANCE****Group A · B · C · D****PYQ FREQUENCY****12+ times (2010–2025)****SOURCE****Caffeine & Cadaver · MBBS PYQ**

★ MBBS PYQ APPEARANCES — YEARS

2010 2010-S 2012-S 2013 2014-S 2015-S 2016 2016-S 2017 2017-S 2018-S 2019
2019-S 2021 2022 2023 2025

📊 EXAM PRIORITY MAP — CARBOHYDRATE METABOLISM

● **TIER 1 — Must-Know (Appear almost every year):** Glycolysis (PFK-1, irreversible steps), TCA Cycle (amphibolic, inhibitors), Glycogen Metabolism (GSD types, hormonal regulation), Gluconeogenesis (bypass enzymes, Cori cycle).

● **TIER 2 — High-Yield (Appear frequently):** PDH Complex (coenzymes, B1 deficiency), HMP Shunt (G6PD deficiency, NADPH functions), Blood Glucose Homeostasis (Fed vs Fasting), Fructose & Galactose disorders.

● **TIER 3 — Low-Yield (Revise only if time permits):** Mnemonics (useful but not directly asked), EQ Answer Template (technique-based).

🔥 00 — LAST DAY REVISION MODE · Ultra-Compressed Protocol

- **Step 1 — Glycolysis:** Net 2 ATP · 3 irreversible steps (HK → PFK-1 → PK) · PFK-1 is RLS · Anaerobic → Lactate (regenerates NAD⁺)
- **Step 2 — PDH:** Pyruvate → Acetyl-CoA · 3 enzymes + 5 coenzymes (TPP/B1, Lipoic acid, CoA/B5, FAD/B2, NAD⁺/B3) · B1 deficiency → Lactic acidosis
- **Step 3 — TCA:** Per turn: 3 NADH + 1 FADH₂ + 1 GTP + 2 CO₂ = 10 ATP · RLS = Citrate Synthase · Amphibolic
- **Step 4 — HMP Shunt:** G6PD is RLS · Products: 2 NADPH + Ribose-5-P · G6PD deficiency → X-linked → Heinz bodies → Haemolysis
- **Step 5 — Glycogen:** Synthase (synthesis) vs Phosphorylase (breakdown) · Glucagon/Epi → cAMP → PKA → breaks glycogen · Insulin = opposite · Von Gierke (Type I) = G6Pase deficiency
- **Step 6 — GNG:** Liver (90%) + Kidney · Precursors = GALA-P · F-2,6-BP is the master switch (Insulin ↑ glycolysis; Glucagon ↑ GNG)

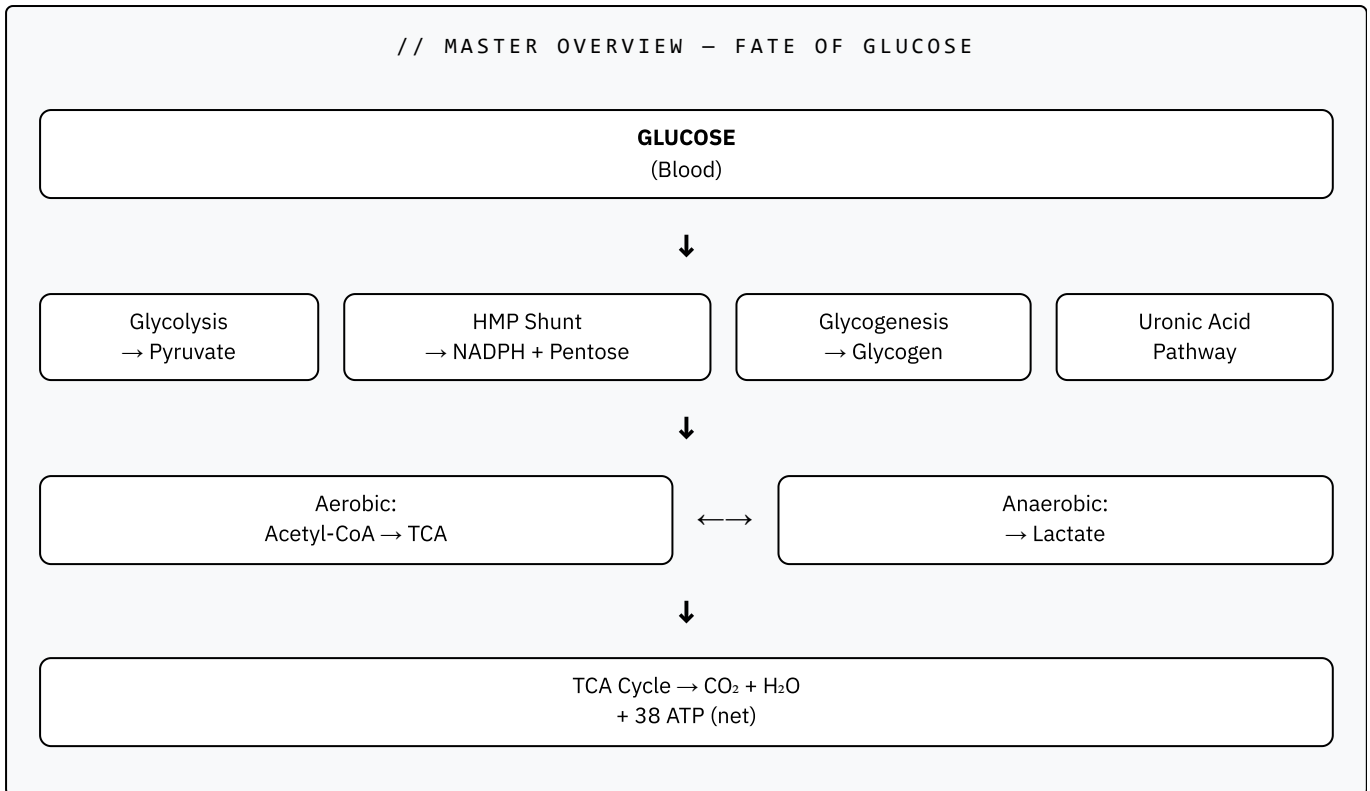
01 The Big Picture

● TIER 2

🔗 WHY THIS MATTERS CLINICALLY

Carbohydrate metabolism is the **central hub of energy production** in the body. Every pathway here connects to a clinical disease — Diabetes (glycolysis regulation), G6PD deficiency (HMP shunt), Von Gierke's disease (glycogen storage), Lactic acidosis (pyruvate dehydrogenase deficiency). Your examiner will test this as Group A almost every year because it demands flowcharts, enzyme knowledge, and clinical correlation all in one answer.

// MASTER OVERVIEW – FATE OF GLUCOSE

02 **Glycolysis**

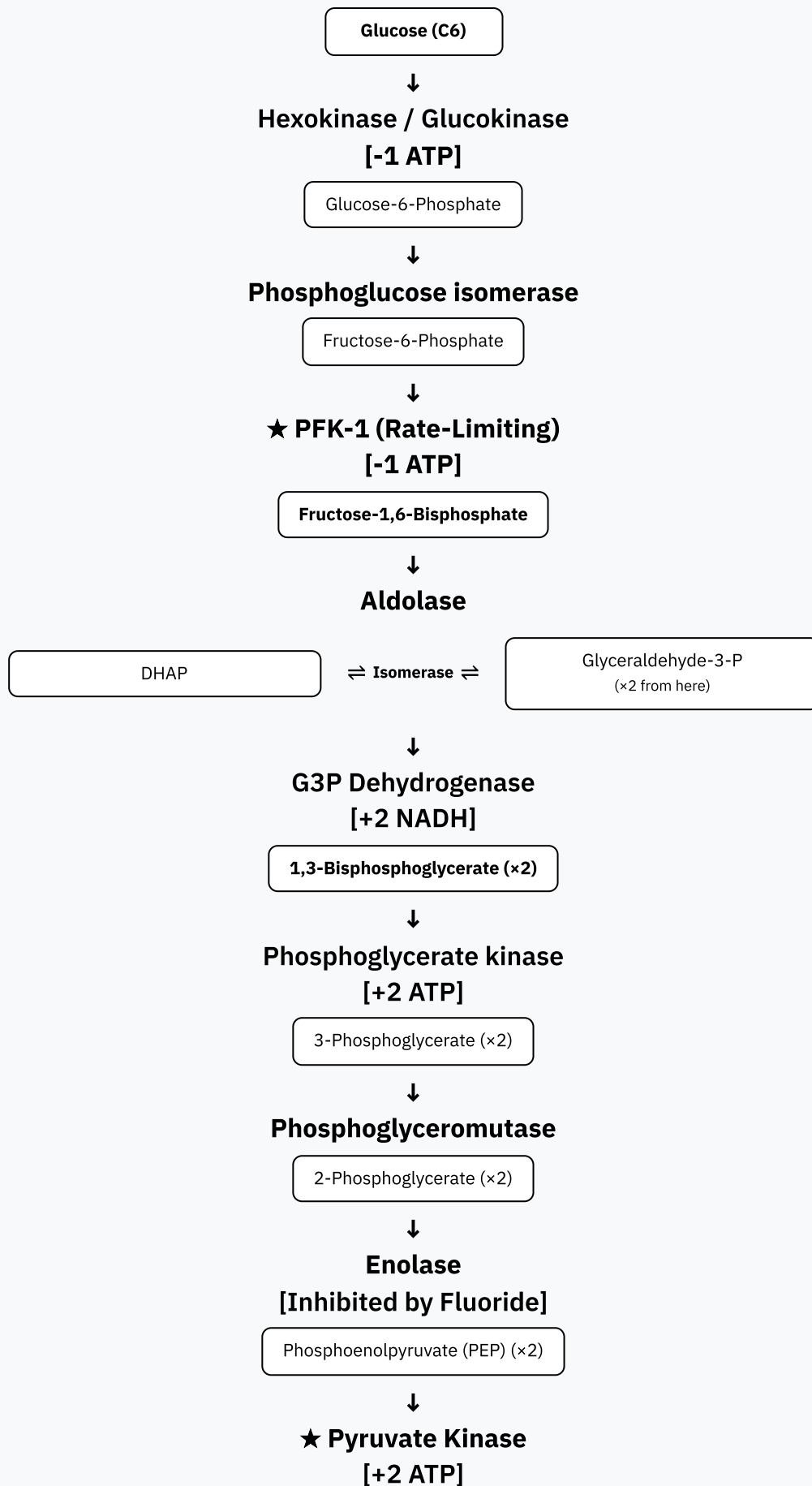
★ EXAMINER FAV

● TIER 1

★ ONE-LINER DEFINITION

Glycolysis is the **cytoplasmic, oxygen-independent** breakdown of one molecule of glucose (6C) into **two molecules of pyruvate (3C)**, yielding a net gain of **2 ATP + 2 NADH**.

// GLYCOLYSIS – 10 STEPS FLOWCHART (EXAM-READY)



Pyruvate (×2) ✓

⚡ ATP invested: 2
ATP gained: 4
Net: +2 ATP

NADH produced: 2
(→ 5 ATP in ETC)

Location: Cytoplasm
O₂ required: X



Rate-Limiting Step:

RATE LIMITING STEP

Phosphofructokinase-1 (PFK-1) — catalyses fructose-6-P → fructose-1,6-bisphosphate. Called the "**pacemaker of glycolysis**." Activated by AMP, ADP, fructose-2,6-bisphosphate. Inhibited by ATP, citrate, low pH.



Anaerobic Glycolysis (in RBCs, muscle during sprint): Pyruvate → Lactate (by LDH). This regenerates NAD⁺ so glycolysis can continue without oxygen. **RBCs have no mitochondria** — 100% dependent on anaerobic glycolysis for ATP.



Rapoport-Luebering Shunt (RBC specific): 1,3-BPG → **2,3-BPG** (bypasses substrate-level phosphorylation → no ATP made here). 2,3-BPG stabilises T-form of Hb → ↓ O₂ affinity → promotes O₂ delivery to tissues.



Clinical Lab Correlation: Fluoride Vials

- Blood for glucose estimation is collected in fluoride vials because Fluoride irreversibly inhibits the enzyme Enolase, stopping RBC glycolysis during transport.

IRREVERSIBLE STEPS OF GLYCOLYSIS (BYPASS POINTS FOR GLUCONEOGENESIS)

Step 1 Glucose → G6P — by **Hexokinase/Glucokinase** (bypassed in GNG by Glucose-6-phosphatase)

Step 3 F6P → F-1,6-BP — by **PFK-1** (bypassed in GNG by Fructose-1,6-bisphosphatase)

Step 10 PEP → Pyruvate — by **Pyruvate Kinase** (bypassed in GNG by Pyruvate carboxylase + PEPCK)



EXAM TRAP — Glycolysis: Students confuse "rate-limiting" with "first step." PFK-1 (Step 3) is the RLS — NOT Hexokinase (Step 1). Also: the net ATP yield is **2 ATP** (not 4) because 2 are invested first. Anaerobic glycolysis ends in **Lactate**, NOT Ethanol (that is yeast fermentation).

MBBS Essay Blueprint: Glycolysis 10-Mark Answering Protocol

If the examiner asks: *"Define glycolysis. Describe its pathway, energetics, and regulation. Add a note on its clinical importance/fluoride vial,"* use this bulletproof structure to secure a perfect 10/10:

1. Definition & Location: Define as the cytoplasmic, oxygen-independent breakdown of glucose to pyruvate (aerobic) or lactate (anaerobic). State it occurs in all cells, localized in the cytosol.

2. The Pathway Flowchart: Draw the 10-step flowchart. Bold the **3 irreversible steps** (Step 1: Hexokinase, Step 3: PFK-1, Step 10: Pyruvate Kinase) using double-line arrows or red asterisks.

3. Energetics Accounting Table: Detail the ATP yield:

PATHWAY STAGE / STEP	AEROBIC GLYCOLYSIS	ANAEROBIC GLYCOLYSIS
ATP INVESTED	-2 ATP (HK & PFK-1 steps)	-2 ATP (HK & PFK-1 steps)
SUBSTRATE-LEVEL PHOSPHORYLATION	+4 ATP (PGK & PK steps ×2)	+4 ATP (PGK & PK steps ×2)
OXIDATIVE PHOSPHORYLATION	+5 ATP (2 NADH × 2.5 via ETC)	0 ATP (NADH consumed by LDH to make lactate)
NET ATP YIELD	+7 ATP (or +5 ATP depending on shuttle)	+2 ATP

4. Key Regulatory Steps: Focus on **PFK-1 (Pacemaker)**. Mention its allosteric activators (AMP, F-2,6-BP) and inhibitors (ATP, Citrate), and touch upon Hexokinase vs. Glucokinase.

5. Clinical Correlations: Briefly explain the mechanism of **Fluoride vials** (Enolase inhibition) and **Rapoport-Luebering Shunt** (RBC specific, 2,3-BPG oxygen delivery).

03 Pyruvate Dehydrogenase Complex

 TIER 2

// PYRUVATE → ACETYL-COA (BRIDGE REACTION)

Pyruvate (3C)

→ CO₂ ↑ + NAD⁺ → NADH

Acetyl-CoA (2C)

PDH COMPLEX – 3 ENZYMES + 5 COENZYMES

E1 – Pyruvate decarboxylase

Coenzyme: **TPP** (Thiamine B1)

E2 – Dihydrolipoyl transacetylase

Coenzyme: **Lipoic acid + CoA**

E3 – Dihydrolipoyl dehydrogenase

Coenzyme: **FAD + NAD⁺**



PDH regulation: Activated by: ADP, CoA, NAD⁺, Ca²⁺, pyruvate. Inhibited by: ATP, Acetyl-CoA, NADH (products inhibit). PDH kinase phosphorylates (inactivates) PDH; PDH phosphatase dephosphorylates (activates) it.



Clinical: Thiamine (B1) deficiency → PDH complex failure → pyruvate accumulates → converted to lactate → **Lactic acidosis**. Presents as Wernicke's encephalopathy in alcoholics. Transketolase (HMP shunt) also needs TPP — used as diagnostic marker.



EXAM TRAP — PDH Complex: PDH needs **5 coenzymes** — TPP (B1), Lipoic acid, CoA (B5), FAD (B2), NAD⁺ (B3). Students often miss Lipoic acid (not a named B-vitamin). Remember: **Arsenite** poisons PDH by inactivating lipoic acid. PDH deficiency causes **Lactic acidosis + neurological symptoms**, NOT ketoacidosis.

04 TCA / Krebs Cycle

★ EXAMINER FAV

● TIER 1



ONE-LINER DEFINITION

The TCA cycle is an **amphibolic** (both anabolic + catabolic), **mitochondrial** cyclic pathway where 1 Acetyl-CoA (2C) condenses with Oxaloacetate (4C) to generate **3 NADH + 1 FADH₂ + 1 GTP + 2 CO₂** per turn.

// TCA CYCLE – COMPLETE FLOWCHART

Oxaloacetate (4C)
+ Acetyl-CoA (2C)

↓ **Citrate Synthase** RLS

Citrate (6C)

↓ **Aconitase**

Isocitrate (6C)

↓ **Isocitrate Dehydrogenase** → CO₂ + NADH

α-Ketoglutarate (5C)

↓ **α-KG Dehydrogenase** → CO₂ + NADH

Succinyl-CoA (4C)

↓ **Succinyl-CoA Synthetase** → GTP ⚡

Succinate (4C)

↓ **Succinate Dehydrogenase** → FADH₂

Fumarate (4C)

↓ **Fumarase** → H₂O added

Malate (4C)

↓ **Malate Dehydrogenase** → NADH

Oxaloacetate (4C) ♻️ regenerated

Per turn:

3 NADH · 1 FADH₂ · 1 GTP · 2 CO₂

ATP yield per turn:

NADH(×3)→7.5 + FADH₂→1.5 + GTP→1
= **10 ATP**

Location:

Mitochondrial matrix

IRREVERSIBLE (REGULATED) STEPS OF TCA

1 Citrate Synthase — Oxaloacetate + Acetyl-CoA → Citrate RLS — Inhibited by: ATP, NADH, Succinyl-CoA, Citrate

3 Isocitrate Dehydrogenase — Activated by: ADP, Ca²⁺. Inhibited by: ATP, NADH

4 α-Ketoglutarate Dehydrogenase — Similar regulation to PDH complex. Needs TPP, Lipoic acid, CoA, FAD, NAD⁺



TCA is AMPHIBOLIC — know why this is asked: Catabolic (breaks down Acetyl-CoA → CO₂ + energy). Anabolic (intermediates drawn off for biosynthesis): α-KG → glutamate → amino acids; Succinyl-CoA → porphyrins/haem; OAA → gluconeogenesis.



TCA Cycle Inhibitors (Frequent PYQ): Fluoroacetate (rat poison) inhibits Aconitase. Malonate is a classic competitive inhibitor of Succinate Dehydrogenase. Arsenite inhibits α-Ketoglutarate dehydrogenase.



EXAM TRAP — TCA Cycle: "Amphibolic" does NOT mean "ambiguous" — it means the cycle serves BOTH catabolic AND anabolic functions. Also: the yield per turn is **10 ATP** (P/O ratio 2.5 for NADH, 1.5 for FADH₂) NOT the old 12 ATP figure. Succinate Dehydrogenase is the ONLY TCA enzyme embedded in the **inner mitochondrial membrane** (all others are in the matrix).



MBBS Essay Blueprint: Amphibolic TCA Cycle 10-Mark Answering Protocol

If the examiner asks: "*TCA cycle is amphibolic in nature. Justify with suitable flowcharts/examples,*" use this high-scoring blueprint to secure 10/10 marks:

1. Definition: Define *amphibolic* as having dual functions—both catabolic (breakdown of Acetyl-CoA to release ATP energy) and anabolic (intermediates drawn off for biosynthetic pathways).

2. Pathway Flowchart: Draw the TCA cycle circle. Draw outward arrows from intermediates to their anabolic fates and inward arrows showing replenishing reactions.

3. The 4 Key Anabolic Draw-off Vectors: Provide this exact table to show your mastery:

INTERMEDIATE	ANABOLIC PATHWAY / FATES
SUCCINYL-COA	Condenses with glycine to form δ-ALA → Heme synthesis.
CITRATE	Exported to cytosol to supply Acetyl-CoA for Fatty Acid & Cholesterol synthesis.
A-KETOGLUTARATE	Transaminated to Glutamate → yields amino acids & GABA neurotransmitter.
OXALOACETATE	Transaminated to Aspartate → Purine / Pyrimidine synthesis, or enters Gluconeogenesis.

4. Justify Anaplerosis (Crucial to secure full marks): Explain that drawing off intermediates would drain the cycle and stop energy production. Explain how **Pyruvate Carboxylase** (biotin-dependent, activated by Acetyl-CoA) replenishes OAA to keep the cycle running.

05 HMP Shunt (Pentose Phosphate Pathway)

● TIER 2

// HMP SHUNT – OXIDATIVE PHASE (HIGH YIELD)

Glucose-6-Phosphate

↓ G6PDH RLS → NADPH + CO₂

6-Phosphogluconolactone

↓ → NADPH

Ribulose-5-Phosphate (5C)

↓ Non-oxidative phase (Transketolase + Transaldolase)

Ribose-5-P
(nucleotides)F-6-P + G-3-P
(re-enters glycolysis)**Products:**2 NADPH · Ribose-5-P · CO₂**Location:**Cytoplasm
Active in: Liver, RBCs,
Adrenal cortex, Testis

Rate-Limiting Step: RATE LIMITING STEP **G6PD (Glucose-6-Phosphate Dehydrogenase)** — generates NADPH. Activated by NADP⁺. Inhibited by NADPH.



Functions of NADPH from HMP shunt: (1) Reduces glutathione (GSH) — protects RBCs from oxidative damage. (2) Required for fatty acid synthesis. (3) Required for cholesterol synthesis. (4) Cytochrome P450 hydroxylation. (5) Phagocyte respiratory burst.

Clinical Vignette Case Study: G6PD Deficiency Hemolytic Crisis

Case Scenario: A 19-year-old male MBBS student presents to the medicine OPD with complains of dark, tea-colored urine, generalized weakness, and mild jaundice. He was recently diagnosed with *Plasmodium vivax* malaria and started on a course of **Primaquine** 48 hours ago.

Lab Findings: Hb: 8.2 g/dL (reduced), Reticulocyte count: 6% (elevated), Peripheral smear shows **Heinz bodies** (aggregated methemoglobin) and "**bite cells**". Indirect bilirubin is elevated.

How to Answer in the Exam:

- **Diagnosis:** G6PD deficiency presenting with acute hemolytic anemia triggered by oxidant drug (Primaquine). (Inheritance: **X-linked recessive**).
- **Why mature RBCs are vulnerable:** RBCs have no mitochondria or ribosomes. They cannot synthesize new proteins. Once G6PD degrades over their lifespan, it cannot be replaced. G6PD is their *only* source of NADPH.
- **Pathophysiology cascade:** Primaquine exposure → generation of reactive oxygen species (ROS) → reduced glutathione (GSH) cannot be regenerated due to lack of NADPH → Methemoglobin precipitates as **Heinz bodies** → splenic macrophages bite out these precipitates, creating **bite cells** and hemolyzing the RBCs.
- **Lab diagnostic caution:** Do *not* measure G6PD activity during the acute hemolytic phase. Active hemolysis clears the most enzyme-deficient (older) RBCs first, leading to a false-normal result. Test G6PD levels 6-8 weeks after the crisis resolves.

06 Glycogen Metabolism

★ EXAMINER FAV

● TIER 1

// GLYCOGENESIS – SYNTHESIS FLOWCHART

Glycogenin: Core protein primer that auto-catalyzes attachment of the first 8 glucose molecules.



Glucose → G6P → G1P

→
+ UTP

UDP-Glucose
(Activated Form)



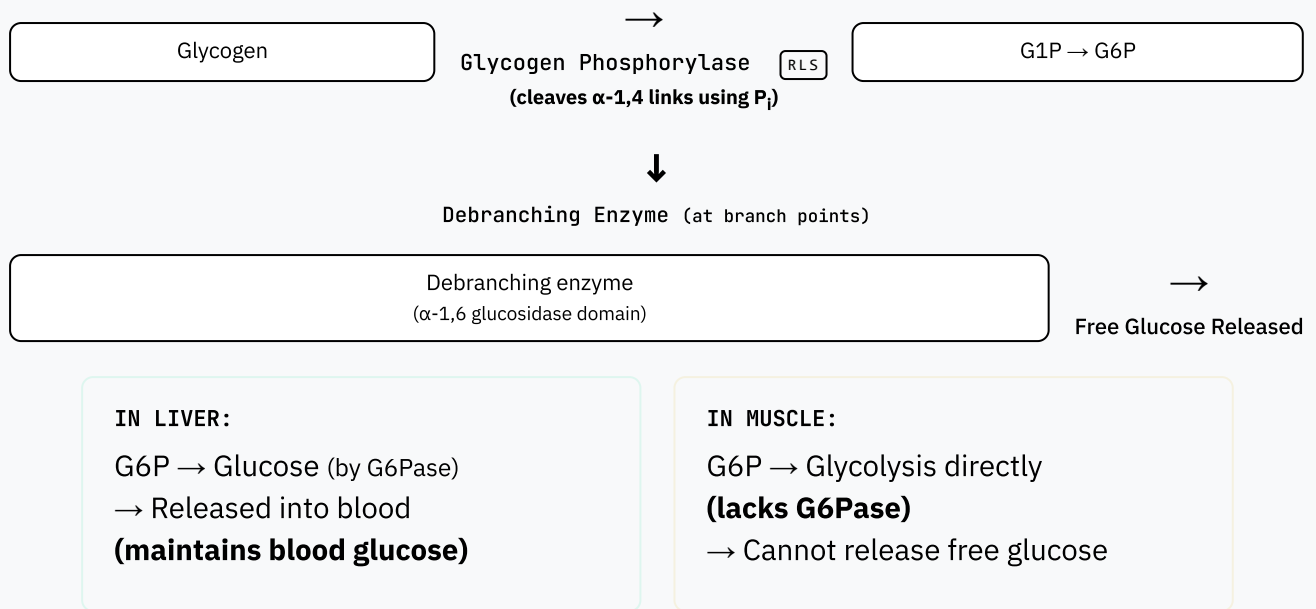
Glycogen Synthase RLS
(adds α-1,4 linkages)

Linear Glycogen Chain

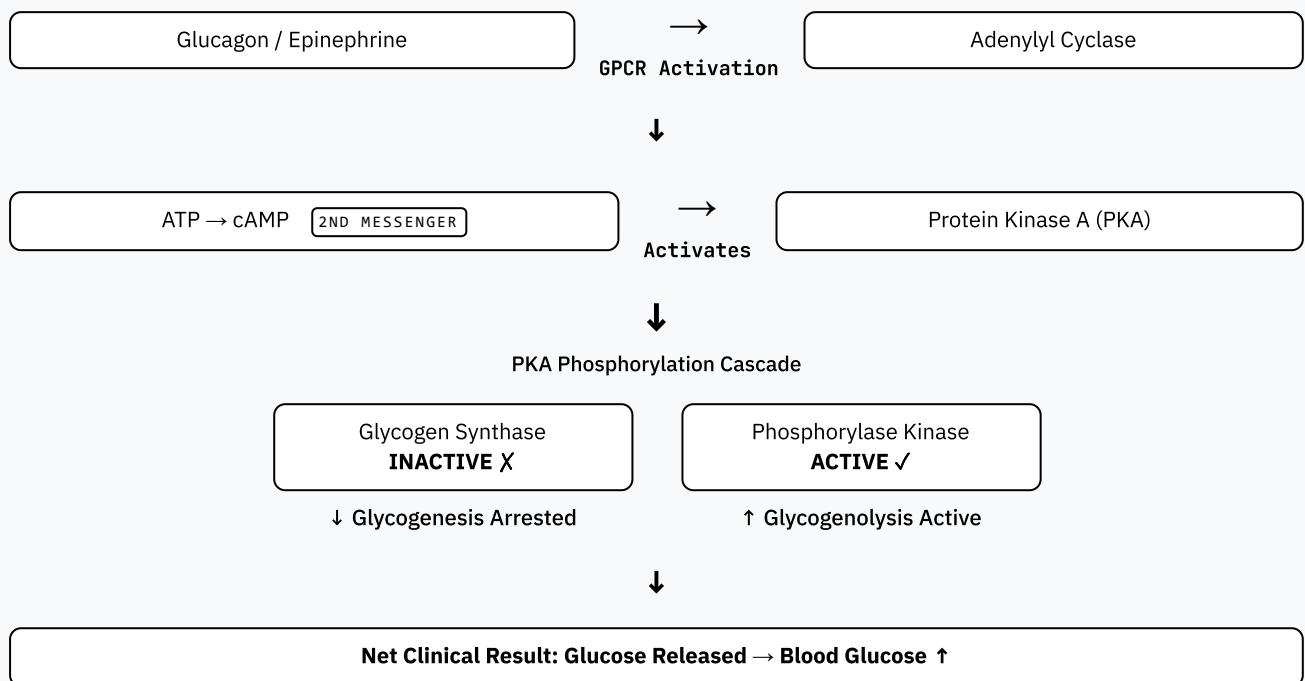
→
Branching Enzyme
(α-1,4 → α-1,6)

Branched Glycogen
(α-1,6 branch points)

// GLYCOGENOLYSIS – BREAKDOWN FLOWCHART



// HORMONAL REGULATION VIA CAMP (GLUCAGON/EPINEPHRINE)



The Insulin Reversal: Activates Protein Phosphatase 1 (PP1) → dephosphorylates both enzymes → activates Glycogen Synthase and inactivates Phosphorylase → promotes Glycogenesis.

HIGH-YIELD GLYCOGEN STORAGE DISEASE (GSD) REVISION GRID

GSD TYPE & NAME	DEFECTIVE ENZYME	KEY SYMPTOMS & CLINICAL SIGN	EXAMINER TRAP / HIGH-YIELD FACT
TYPE I: VON GIERKE'S	Glucose-6-phosphatase (G6Pase)	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis.	Hyperuricaemia (Gout) due to shunting of accumulated G6P into the HMP pathway.
TYPE II: POMPE'S	Lysosomal α -1,4-glucosidase (Acid Maltase)	Massive cardiomegaly , generalized hypotonia.	Lysosomal storage disease (glycogen builds in vacuoles). Blood glucose is normal.
TYPE III: CORI'S	Debranching enzyme (α -1,6-glucosidase)	Milder fasting hypoglycemia, hepatomegaly.	Accumulation of limit dextrins (abnormally short outer branches).
TYPE V: MCARDLE'S	Muscle Glycogen Phosphorylase	Exercise-induced cramps, myoglobinuria (red urine).	No rise in blood lactate after exercise (ischemic exercise test is diagnostic).

07 Gluconeogenesis

● TIER 1

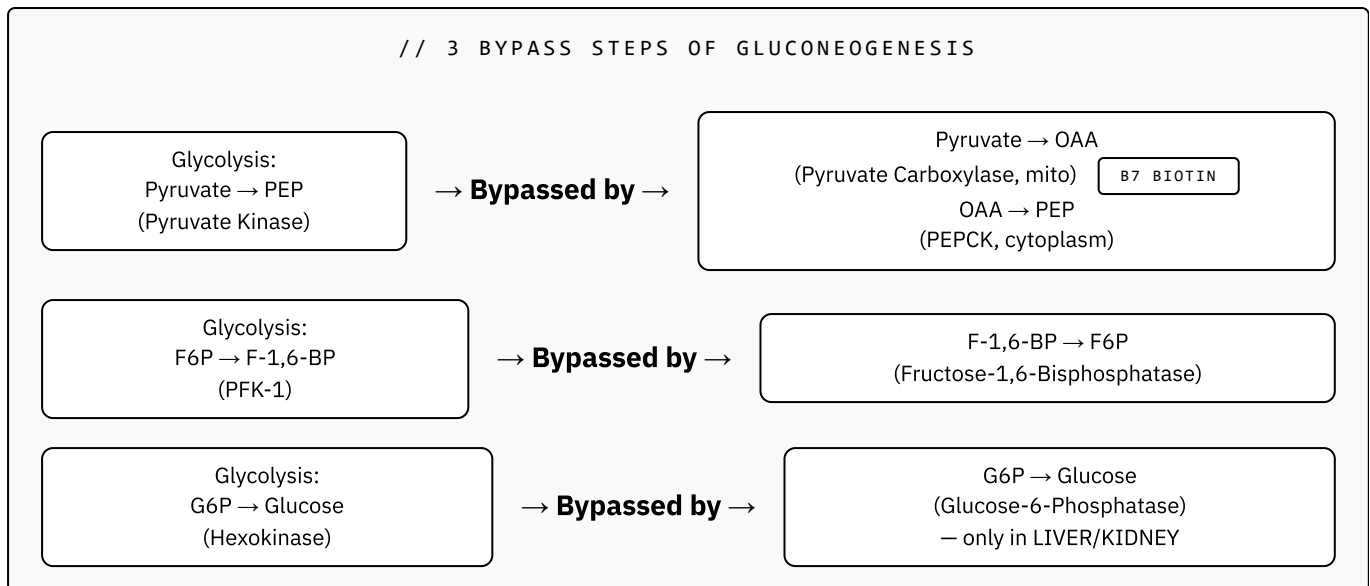
★ DEFINITION

Synthesis of **new glucose from non-carbohydrate precursors**. Occurs mainly in **liver (90%)** and kidney cortex (10%) during fasting/starvation. Not simply the reverse of glycolysis — three irreversible steps of glycolysis are bypassed by unique enzymes.



Precursors (GALA-P mnemonic): Glycerol (from TAG breakdown) · Alanine + other amino acids (glucogenic AAs) · Lactate (Cori cycle) · Acetyl-CoA (indirectly, via OAA) · Propionate (odd-chain FA → Succinyl-CoA)

// 3 BYPASS STEPS OF GLUCONEOGENESIS



Cori Cycle (Lactic acid cycle): Muscle lactate → blood → liver → gluconeogenesis → glucose → blood → back to muscle. Net: transfers metabolic burden from muscle to liver. Energy-expensive: It costs exactly 6 ATP equivalents (4 ATP + 2 GTP) in the liver to synthesize 1 molecule of glucose from 2 pyruvates, compared to the 2 ATP gained in muscle glycolysis.



Regulation by Fructose-2,6-Bisphosphate (F-2,6-BP): CLASSIC PYQ 2017 F-2,6-BP activates PFK-1 (glycolysis) and inhibits F-1,6-BPase (GNG) — a single molecule that switches between the two pathways. Insulin ↑ F-2,6-BP → glycolysis. Glucagon ↓ F-2,6-BP → GNG.



EXAM TRAP — Gluconeogenesis: Students often write that Gluconeogenesis occurs in the muscle. This is a fatal mistake! **Muscle cannot perform gluconeogenesis** because it lacks the key enzyme **Glucose-6-Phosphatase**. Also, fatty acids (even-chain) **cannot** serve as a precursor for glucose in humans because the Pyruvate Dehydrogenase reaction is irreversible (Acetyl-CoA cannot be converted back to pyruvate).



MBBS Essay Blueprint: Gluconeogenesis & F-2,6-BP Switch Answering Protocol

If the examiner asks: *"Define Gluconeogenesis. State its substrates, describe the bypass reactions, energy costs, and the reciprocal regulation by F-2,6-BP switch,"* organize your answer using this textbook structure to score maximum marks:

1. Definition & Tissue Location: Synthesis of glucose from non-carbohydrate precursors. Location: primarily **Liver (90%)** and **Kidney cortex (10%)** (NOT muscle!). Subcellular site: compartmentalized between **Mitochondria, Cytosol, and Endoplasmic Reticulum**.

2. Precursors List (GALA-P): Glycerol (from lipid lipolysis), Amino acids (primarily Alanine/Gluconeogenic AAs), Lactate (from Cori cycle), Propionate (from odd-chain fatty acids).

3. The 3 Compartmentalized Bypass Reactions:

BYPASS STEP	GLYCOLYTIC ENZYME (IRREVERSIBLE)	GLUCONEOGENIC BYPASS ENZYMES & COMPARTMENT
BYPASS 1	Pyruvate Kinase	1. Pyruvate Carboxylase (Mitochondria; requires Biotin, ATP, CO ₂ ; activated by Acetyl-CoA) 2. PEPCK (Cytosol; converts OAA to PEP; requires GTP)
BYPASS 2	Phosphofructokinase-1 (PFK-1)	Fructose-1,6-Bisphosphatase (Cytosol; allosterically inhibited by F-2,6-BP & AMP)
BYPASS 3	Hexokinase / Glucokinase	Glucose-6-Phosphatase (Endoplasmic Reticulum membrane; absent in skeletal muscle)

4. Energetic Cost Accounting: To synthesize **1 mole of glucose from 2 moles of Pyruvate**, it costs exactly **6 high-energy phosphate bonds**:

- 2 ATP invested at the *Pyruvate Carboxylase* reaction (1 per Pyruvate)
- 2 GTP invested at the *PEPCK* reaction (1 per OAA)
- 2 ATP invested at the *1,3-BPG kinase* step (reversed)
- *Total cost = 4 ATP + 2 GTP = 6 High-Energy Bonds* (compared to only 2 ATP produced in glycolysis).

5. The Reciprocal F-2,6-BP Master Switch: Describe how the bifunctional enzyme **PFK-2 / FBPase-2** governs the switch:

- **Fed State / High Insulin:** Dephosphorylation activates the PFK-2 domain → ↑ F-2,6-BP levels → powerfully activates PFK-1 (stimulates Glycolysis) & inhibits F-1,6-BPase (shuts down GNG).
- **Fasting State / High Glucagon:** GPCR-cAMP-PKA pathway phosphorylates the bifunctional enzyme, activating the FBPase-2 domain → ↓ F-2,6-BP levels → relieves inhibition on F-1,6-BPase (stimulates GNG) & shuts off Glycolysis.

08 Blood Glucose Homeostasis

TIER 2

// FED VS FASTING STATE

FED STATE (↑ Insulin)

- Glucose uptake ↑ (GLUT4 in muscle/fat)
- Glycolysis ↑
- Glycogenesis ↑
- Lipogenesis ↑
- Gluconeogenesis ↓
- Protein synthesis ↑

FASTING STATE (↑ Glucagon)

- Glycogenolysis ↑ (liver first)
- Gluconeogenesis ↑ (after 4–6h)
- Lipolysis ↑ → FFA + Ketones
- Protein catabolism ↑ (AA for GNG)
- Glycolysis ↓
- Glycogenesis ↓

09 Fructose & Galactose Metabolism

TIER 2



FRUCTOSE DISORDERS: ESSENTIAL FRUCTOSURIA VS. HEREDITARY FRUCTOSE INTOLERANCE (HFI)

FEATURE	ESSENTIAL FRUCTOSURIA	HEREDITARY FRUCTOSE INTOLERANCE (HFI) CRITICAL
DEFICIENT ENZYME	Fructokinase (first step of fructose metabolism)	Aldolase B (cleaves Fructose-1-Phosphate)
ACCUMULATION	Fructose (excreted in urine; not toxic inside cells)	Fructose-1-Phosphate (F-1-P) (trapped inside hepatocytes)
PATHOPHYSIOLOGY	Fructose is not phosphorylated; benignly filtered by kidneys or bypassed via Hexokinase. No ATP depletion.	Phosphate Trapping: Buildup of F-1-P sequester-depletes intracellular inorganic phosphate (P_i) and ATP . Lack of ATP/P _i halts Glycogenolysis & Gluconeogenesis!
CLINICAL SIGNS	Asymptomatic (incidentally detected as a reducing sugar in urine; Clinitest positive, glucose stick negative).	Severe hypoglycemia, vomiting, jaundice, hepatomegaly, lactic acidosis, proximal renal tubule dysfunction (Fanconi syndrome) upon fructose/sucrose ingestion.
DIETARY THERAPY	No restriction required. Benign condition.	Strict elimination of Fructose, Sucrose, and Sorbitol from diet.
EXAMINER TRAP	Fructose gives a positive reducing sugar test (Clinitest). Do not confuse with Diabetes Mellitus (glucose dipstick is negative!).	Symptoms appear only after weaning (introduction of fruit juices, honey, or formula milk containing sucrose).



GALACTOSE DISORDERS: GALK DEFICIENCY VS. CLASSICAL GALACTOSEMIA (GALT DEFICIENCY)

FEATURE	GALACTOKINASE (GALK) DEFICIENCY	CLASSICAL GALACTOSEMIA (GALT DEFICIENCY)
		CRITICAL
DEFICIENT ENZYME	Galactokinase (GALK) (phosphorylates galactose)	Galactose-1-Phosphate Uridyltransferase (GALT)
ACCUMULATION	Galactose (excreted in urine) & Galactitol (reduced by Aldose Reductase)	Galactose-1-Phosphate & Galactitol (highly toxic)
PATHOPHYSIOLOGY	Galactose accumulates and is shunted to Galactitol. Galactitol accumulates in the lens, causing osmotic damage.	Galactose-1-P accumulation traps inorganic phosphate and acts as a direct tissue toxin to liver, brain, kidneys, and lens.
CLINICAL SIGNS	Mild presentation: Infantile cataracts, galactosemia, galactosuria. No renal/liver failure or developmental delay.	Severe presentation: Failure to thrive, jaundice, hepatomegaly, infantile cataracts, intellectual disability, and life-threatening E. coli neonatal sepsis .
DIETARY THERAPY	Eliminate galactose and lactose (milk/dairy products) from diet.	Strict elimination of Galactose and Lactose (human and bovine milk) immediately from birth.
EXAMINER TRAP	Only cataracts are present. No liver damage or developmental delay. Contrast this with classical galactosemia.	Symptoms start within the first few days of life, as soon as breastfeeding/milk feeding begins (unlike HFI, which starts later with weaning!).

10 **Memory Mnemonics**

● TIER 3

● TIER 3 – Low-Yield. Click to expand Mnemonics →

● MNEMONIC 1 – TCA INTERMEDIATES IN ORDER

"Citrate Is Krebs' Starting Substrate For Making Oxaloacetate"Citrate → Isocitrate → α -Ketoglutarate → Succinyl-CoA → Succinate → Fumarate → Malate → Oxaloacetate

● MNEMONIC 2 – IRREVERSIBLE GLYCOLYSIS STEPS

"Hungry People Feel Pale" → HK, PFK-1, PK

Hexokinase (step 1) · Phosphofructokinase-1 (step 3) · Pyruvate Kinase (step 10) — these 3 are irreversible and are bypassed in gluconeogenesis.

● MNEMONIC 3 – PDH COMPLEX COENZYMES

"The Lovely Cat Finds Nice Accommodation"TPP (B1) · Lipoic acid · CoA (B5 pantothenic acid) · FAD (B2) · NAD⁺ (B3) · Note: All B-vitamins!

● MNEMONIC 4 – GLYCOGEN STORAGE DISEASES

"Very Poor Carbohydrate Metabolism" → Types I, II, III, VType I = **V**on Gierke (G6Pase) · Type II = **P**ompe (Acid maltase) · Type III = **C**ori (Debranching) · Type V = **M**cArdle (Muscle phosphorylase)

● MNEMONIC 5 – GNG PRECURSORS

"GALA Party" → Glycerol, Amino acids, Lactate, Alanine / Propionate

All glucogenic amino acids → glucose. Fatty acids (even-chain) CANNOT → glucose (no net conversion from Acetyl-CoA). Odd-chain FA → Propionyl-CoA → Succinyl-CoA → CAN enter GNG.

11 Clinical Correlations

● TIER 2

 High-Yield Clinical Links

- **Diabetes Mellitus:** Insulin deficiency → ↓ GLUT4 → ↓ glucose uptake → Hyperglycaemia. ↑ Glycogenolysis, ↑ GNG, ↑ Lipolysis, ↑ Ketogenesis → Diabetic Ketoacidosis (DKA). HbA1c reflects average glucose over 3 months (glycated Hb).
- **Lactic Acidosis:** Anaerobic glycolysis excess (shock, sepsis) or PDH deficiency → Lactate accumulates → ↓ pH. High Anion Gap Metabolic Acidosis.
- **Von Gierke's + Hyperuricaemia:** ↑ G6P → ↑ HMP shunt → ↑ Ribose-5-P → ↑ Purine synthesis → ↑ Uric acid. Classic PYQ trap — explain the link!
- **Galactosaemia → Cataract:** Galactose → Galactitol (by aldose reductase) → Accumulates in lens (avascular) → Osmotic damage → Cataract. Treat: eliminate galactose/lactose from diet.
- **Uncontrolled DM → Cataract:** Glucose → Sorbitol (aldose reductase) → Accumulates in lens → Same mechanism as galactosaemia.
- **Primaquine + G6PD deficiency:** Oxidative drug → ↑ ROS → ↓ NADPH (G6PD absent) → ↓ GSH → Heinz bodies → Haemolysis. Tested in clinical MCQs.

 VIVA RAPID FIRE – CARBOHYDRATE METABOLISM**Q1 Why can't RBCs use the TCA cycle?**

RBCs have no mitochondria — TCA cycle and oxidative phosphorylation are mitochondria-dependent. RBCs rely entirely on anaerobic glycolysis for ATP. They also use HMP shunt to generate NADPH for maintaining reduced glutathione (protection against oxidative damage).

Q2 Why does Von Gierke's disease cause hyperuricaemia?

G6Pase deficiency → G6P accumulates → excess G6P shunted into HMP pathway → excess Ribose-5-phosphate → increased de novo purine synthesis → excess purines → increased uric acid production → hyperuricaemia and gout.

Q3 Why is gluconeogenesis NOT the simple reverse of glycolysis?

Three irreversible steps of glycolysis cannot be reversed. These are catalysed by Hexokinase, PFK-1, and Pyruvate Kinase. GNG uses unique bypass enzymes: Glucose-6-phosphatase, Fructose-1,6-bisphosphatase, and the Pyruvate Carboxylase + PEPCK system respectively.

Q4 What is the role of 2,3-BPG? Why is it important in stored blood?

2,3-BPG binds to the β -chains of deoxy-Hb, stabilising the T (tense/deoxy) form, reducing O_2 affinity → promotes O_2 delivery to tissues. In stored blood (bank blood), 2,3-BPG levels fall → Hb has increased O_2 affinity → poor O_2 delivery to recipient's tissues (stored blood paradox).

Q5 Why is TCA called amphibolic?

Amphibolic = both anabolic and catabolic. Catabolic: oxidises Acetyl-CoA → CO_2 + energy. Anabolic: intermediates are precursors for biosynthesis — α -KG → glutamate → amino acids; Succinyl-CoA → haem synthesis; OAA → gluconeogenesis; Citrate → fatty acid synthesis.

Q6 Why does muscle lack glucose-6-phosphatase but liver has it?

Liver's role is to maintain blood glucose homeostasis — it must release free glucose into circulation during fasting. G6Pase allows dephosphorylation of G6P → free glucose → export. Muscle is a glucose consumer — it needs G6P to stay within the cell for glycolysis. If muscle had G6Pase, glucose would leak out rather than be used for energy.

Q7 Fat can be synthesised from glucose but glucose cannot be synthesised from fat — justify.

Glucose → Acetyl-CoA (irreversible via PDH). Fatty acids → β -oxidation → Acetyl-CoA. Acetyl-CoA cannot be converted back to pyruvate/OAA in net amounts in mammals — the PDH reaction is irreversible. Acetyl-CoA can only enter TCA (catabolic) or be used for lipid synthesis, not for GNG. Exception: glycerol backbone of TAG can yield glucose (via DHAP).

Q8 What is the significance of oral glucose in ORS (Oral Rehydration Solution)?

Glucose in ORS co-transporters sodium via SGLT1 (Na^+ -glucose co-transporter) in the intestinal mucosa — a secondary active transport system. This Na^+ uptake drags water with it by osmosis, rehydrating the patient. Glucose doesn't just provide energy — its primary role in ORS is to drive Na^+ absorption even in secretory diarrhoea where other Na^+ transporters are inhibited.

13 **Group-D EQ Template**

● TIER 3

● TIER 3 – Low-Yield. Click to expand EQ Template →

HOW TO ANSWER: "PHOSPHOFRUCTOKINASE-I IS THE PACEMAKER OF GLYCOLYSIS"

Use your teacher's flowchart formula — Define → Mechanism Flowchart → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART

Define "Pacemaker"

Rate-limiting, essentially irreversible, most regulated step

**Reaction**

$$\text{F6P} + \text{ATP} \rightarrow \text{F-1,6-BP} + \text{ADP}$$
 (catalysed by PFK-1)
**Activators**AMP, ADP, F-2,6-BP, Pi
(signals: low energy)**Inhibitors**ATP, Citrate, low pH
(signals: high energy)**Clinical relevance**F-2,6-BP (insulin ↑, glucagon ↓)
coordinates glycolysis vs GNG**MBBS Biochemistry – Module 02: Carbohydrate Metabolism**

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